The Effects and Dynamics of Blind Hiring

By

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Abstract

This paper investigates the dynamics of labor market discrimination and the efficacy of blind hiring across multiple stages of the hiring process. We conduct a two-phase experiment in the U.S. technology sector to examine how name-implied demographic cues influence evaluations by resume screeners and technical interviewers involved in different hiring stages. We find that evaluators apply different decision rules depending on their role: whereas HR professionals exhibit little systematic demographic differentiation, engineers assign significant positive premia to several non-White minority groups during the technical evaluation stage. Consequently, removing names through blind hiring mechanically reverses these patterns, penalizing previously favored minority applicants and boosting ratings for White male candidates. By mapping these evaluations into a two-stage hiring process, we show that the impact of blind hiring on the final demographic composition of hires depends heavily on the inclusivity of the initial resume screen. Furthermore, while anonymizing only the resume-screening stage does not meaningfully affect hire quality, blinding both the resume-screening and technical-interviewing stages systematically reduces the objectively measured coding ability of selected candidates. These findings highlight the complex, sometimes unintended consequences of removing demographic signals in a multi-stage hiring pipeline.

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I Introduction

The discrimination faced by workers in the labor market has been widely documented in previous research, usually through audit or correspondence studies (Bertrand and Mullainathan, 2004; Kroft et al., 2013; Bertrand and Duflo, 2017; Agan and Starr, 2018; Neumark, 2021; Kessler et al., 2019; Kline et al., 2022). However, many of the articles studying discrimination focus primarily on discrimination that occurs in the *resume screening stage* of the job application rather than in the *interview stage*, leaving a gap in measuring discrimination during the job hiring process. This paper examines the dynamics of discrimination across both the resume-screening and interview stages, and evaluates blind hiring as a potential policy intervention to mitigate such discrimination.

Blind hiring is used as a policy tool to combat labor market discrimination by preventing employers from observing the' group identity of job applicants (Goldin and Rouse, 2000; Åslund and Nordström Skans, 2012; Lee et al., 2018; Min and Son, 2020; Neumark, 2021; Park, 2021).[1] However, in practice, most employers can observe the group identity of applicants during the interview stage, even if the name is concealed on the resume. Thus, any effort to study the effect of blind hiring when there is an interview stage is likely to overestimate the effect of blind hiring.

In this paper, in the context of the U.S. tech industry, we address three main questions: (i) do decision-makers at different stages of hiring apply different decision rules based on the demographic identities of the candidates?; (ii) how does blinding the observable group characteristics affect employers' subjective evaluation of job applicants? and (iii) how does blind hiring interact with multiple stages of hiring and alter the composition or productivity of candidates who are ultimately hired?

To answer the above questions, we implement a two-phase experiment, consisting of an applicant phase and an employer phase, that recreates a stylized the hiring funnel in the U.S. technology sector, separating HR's applicant screening from software engineer's technical evaluations. In the applicant phase, we recruit UC Berkeley undergraduate students majoring in Computer Science or EECS to participate as job applicants in a simulated hiring process. Students attend a lab session to complete a standardized coding assessment designed to resemble the technical screens used for entry-level software engineering roles and submit their most recent resume. We construct a measure of coding ability based on their lab performance using evaluations by two professional software engineers. This index serves as a proxy for applicant productivity within the simulated hiring environment.

In the employer phase, HR professionals and software engineers in the U.S. technology sector evaluated these applications using an incentivized resume-rating method. Each evaluator reviews a fixed set of applications (18 in our implementation). HR professionals evaluate only the resume, corresponding to the resume-screening stage, while software engineers evaluate the same candidates using both the resume and the coding lab performances, corresponding to a technical evaluation stage. To incentivize truthful and effortful ratings, evaluators are promised access to a

[1] For example, the South Korean government implemented a blind hiring policy for all public institution job postings, where employers are not allowed to observe the name, gender, age and university name;

subset of real UC Berkeley candidates whose resumes best match the preferences they reveal in the task.

Each HR and Engineer participants were randomly assigned to either a *Nonblind* condition in which applicants' first and last names are shown or a *Blind* condition in which names are removed. The names were selected to convey commonly perceived race and gender associations, and we ensure that no other resume content reveals these characteristics, so treatment (blinding) isolates access to demographic signals at each stage. This structure allows us to estimate stage-specific effects of demographic signals (and blinding thereof) on job applicant evaluations and to trace how blinding reshapes the demographic composition and expected ability of candidates who advance.

Our results show that name-implied demographic cues shape employer evaluations, and that the importance of these cues varies across hiring stages. In the non-blind condition – where evaluators observe applicant names – HR professionals exhibit little systematic differentiation in ratings across demographic groups, with one notable exception: resumes assigned Hispanic male names receive modestly higher assessments on predicted coding ability and overall evaluations relative to identical resumes assigned White male names. By contrast, software engineers display more pronounced name-based differences. Several non-White groups receive positive "premia" on key outcomes, such as predicted coding ability, workplace fit, and overall ratings, indicating that demographic cues conveyed through names can influence assessments even at a more technical, information-rich stage of hiring.

Turning to the causal effects of blind hiring, we find that removing names changes evaluations in ways that closely mirror the premia and penalties observed when names are visible. Among HR professionals, blinding substantially lowers evaluations for resumes assigned Hispanic male names, while effects for other groups are comparatively small and less systematic. Among engineers, blinding tends to increase ratings for White male candidates and reduce ratings for demographic groups that previously received positive name-based premia under non-blind review (notably Asian male and Hispanic female candidates on key dimensions). Overall, the pattern is consistent with blinding operating primarily by removing demographic cues that were affecting subjective assessments, rather than by altering anything about resume content.

We then study how these shifts propagate through a two-stage hiring process by pairing HR and engineer evaluations and mapping them into stage-specific selection rules. Here, the effects of complete blind hiring (blinding at both stages) depend on how selective the HR stage is and how many finalists the engineer chooses. When HR screening is relatively strict (advancing only a small top set), the effects of blinding on which demographic groups are selected are generally muted. When HR screening is more inclusive (advancing a larger pool), on the other hand, blinding has more room to shift the composition of selected candidates, with changes becoming more pronounced as engineers select more finalists.

Finally, we examine whether blind hiring affects the quality of the ultimately selected candidate as measured by a pre-determined coding assessment score. Blinding only at the HR stage (a "realistic" blind hiring policy where engineers still see names) has little effect on the coding abil-

ity of the selected candidate. In contrast, complete blind hiring tends to reduce the coding score of the selected candidate across alternative funnel definitions and selection rules. These patterns should not be interpreted as evidence that names contain valid information about ability. Rather, our findings highlight that removing demographic cues can alter evaluators' rankings and candidate selection in ways that may affect measured outcomes – even when the removed cue is, by construction, not informative about true skill.

The remainder of the paper is structured as follows: Section II details the experimental design; Section IV outlines the empirical strategy and presents preliminary results; and the final Section V concludes.

II Experimental Design

To simulate the standard hiring process for software engineers, we implemented a two-phase experimental design consisting of an applicant phase and an employer phase. In the applicant phase, we collect the inputs needed to construct standardized job-applicant profiles: student resume and performance on a coding assessment. In the employer phase, we mimic a two-stage hiring funnel by having HR professionals evaluate resume (*resume screening stage*) and software engineers evaluate the same candidates using both resume and coding performance (*technical evaluation stage, analogous to a coding interview*). For both types of evaluators, we randomly vary whether applicants' demographic (race and gender) signals are visible (Non-blind / Control) or redacted (Blind / Treatment), allowing us to estimate stage-specific effects of blinding demographic signals on evaluations.

II.A Job Applicant Phase

The applicant phase is designed to generate standardized job-applicant profiles for the employer-side evaluation. Each participating student submits an up-to-date resume and completes a proctored coding assessment designed to mirror technical evaluations commonly used by technology firms. Participants also complete a brief survey that records demographic characteristics and elicits perceptions of discrimination in the tech labor market.

One strength of our design is that the coding assessment yields an ex ante measure of coding ability – a predetermined proxy for productivity – for every job applicant. Because the assessment is administered under standardized, proctored settings and graded independently by external software engineers using a pre-specified rubric, the resulting coding-ability index is determined prior to the employer phase and is therefore invariant to employer-side experiment nor treatment assignment. This feature, among many things, allows us to quantify how blinding impacts the expected ability of candidates who are selected.

Recruitment and incentives. We invited third- and fourth-year undergraduates majoring in Computer Science (CS) or Electrical Engineering and Computer Sciences (EECS) at the University of

California, Berkeley (UCB) who were seeking employment in software-engineering roles. We recruited 69 participants through an email invitation distributed by the department. We incentivized full-effort participation in two ways. First, students received $20 for participation and were eligible for a performance bonus of up to $20 based on their coding-lab performance.[2] Second, students were informed that their resumes and coding submissions could be shared with employers participating in the employer phase, who could contact them regarding referrals or potential job opportunities.[3] Students were free to ignore any employer outreach if they were not interested.

Coding assessment and grading. During the last two weeks of September 2025, we administered six in-person coding-lab sessions at the Xlab at UC Berkeley. Each student attempted two coding questions in 60 minutes using an online coding interface hosted on a professional assessment platform.[4][5] To ensure assessment integrity, lab computers were configured to restrict access to external resources (e.g., through Firefox kiosk-mode and key-blockers), and all sessions were proctored. Participants could code in C++, Python, or Java.[6] The coding questions are provided in Appendix C.I.2.

Coding submissions were graded anonymously by two independent professionals with substantial industry experience using a pre-specified rubric (See Section II.C for details).

Survey and resume standardization. In addition to completing the coding assessment, students submitted their most recent resumes and completed a short survey eliciting demographics and perceived discrimination in the tech industry. We reformatted all resumes into a standardized template for use in the employer-side experiment. To isolate the role of name-based demographic cues, we edited resumes to remove other salient race- and gender-related signals while making minimal changes to substantive content. For example, when a leadership entry explicitly referenced a women-only organization, we made minimal wording edits to eliminate the gender-specific identifier while preserving the underlying experience. Researchers and research assistants reviewed re-formatted resumes manually (with additional automated checks using AI) to ensure that non-name demographic cues were not visible. A small number of resumes containing prominent demographic signals that could not be credibly removed without materially altering content were excluded from the employer-side experiment.

II.B Employer Side

In the employer-side experiment, we ask employers to evaluate standardized applicant profiles (Section II.A) to study whether evaluations change when different demographic signals are pre-

[2]The average performance-based bonus paid was $14.

[3]Students could later opt out of having their resumes and/or coding submissions shared with employers; 7 students chose not to share.

[4]Remote Interview IO is commonly used in industry to conduct technical screening and live coding interviews.

[5]Coding questions were taken from LeetCode and modified under the guidance of software engineers with industry experience.

[6]These are commonly used languages among UC Berkeley computer science students.

sented or redacted (blinded). The experiment was implemented as an online survey administered in Qualtrics. To mirror common hiring practices in the U.S. technology sector, we recruited two types of decision-makers: Human Resources (HR) professionals, who typically conduct initial resume screening, and software engineers, who assess candidates' technical ability in subsequent evaluations.[7]

Recruitment. We recruited employer participants from the U.S. I.T. industry who reported having experience recruiting for entry-level software engineer roles. The employer-side survey was fielded between October 30, 2025 and January 21, 2026. Using a commercial contacts database,[8] we sent 73,555 email invitations.[9] In total, 582 individuals (0.79% of contacted recipients) completed the survey, including some respondents who were subsequently screened out due to ineligibility (i.e. foreign IP address); our final analytic sample consists of 150 HR professionals and 170 software engineers. At the end of the survey, participants were invited to share the study with colleagues; 18 participants (12 HR and 6 engineers) were enrolled via peer invitation.

Evaluation task. Each employer participant evaluated the same set of 18 applicant profiles drawn from our student pool.[10] HR professionals reviewed a uniformly formatted resume only, corresponding to the *resume screening stage*. (See Figure C.3 for an example of HR's candidate-evaluation survey page.) Software engineers evaluated the same applicants but observed both the formatted resume and the applicant's coding submission from the proctored evaluation, corresponding to the *technical evaluation stage*. (See Figure C.5 for an example of engineers' candidate-evaluation survey page.) Engineers were additionally shown objective execution information, including whether the code compiled and how many of the pre-specified test cases the code correctly executes. Aside from the experimental manipulation of name-based demographic cues, all resume formatting and non-name content were held fixed across applicants and across experimental conditions.

Both HR and engineer participants rated each candidate on (i) an overall score on a 0–100 scale and (ii) three sub-dimensions—coding ability, workplace fit, and likelihood of staying for five years—on 0–10 scales. In addition, HR participants were asked whether the candidate should be shortlisted to advance to the next *interview* stage for technical evaluation by an engineer.

Randomization of blind treatment and applicant names. Within each evaluator type (HR and engineer), we randomly assigned participants to either a *Non-blind* (control) condition or a *Blind* (treatment) condition. In the Non-blind condition, resumes displayed applicants' hypothetical

[7] In many technology firms, HR conducts an initial resume screen to determine which applicants advance, while engineers assess technical skills through coding-based evaluations on a small subset of shortlisted candidates.

[8] Industry contacts were collected via Apollo.io, which allows targeting by firms' industry and by individuals' job title/department and location. The platform also provides LinkedIn profile links, which we used—in addition to voluntarily self-provided LinkedIn profiles—to verify respondent identities and roles.

[9] This count excludes invitation emails that bounced and therefore did not reach the intended recipients.

[10] From the 69 student profiles collected in the Job Applicant Phase, we first removed a small number of resumes that contained strong demographic signals in components that could not be removed. We then randomly selected 18 profiles, stratifying by demographic group and baseline academic/performance measures (GPA and coding score).

names; in the Blind condition, names were removed so that name-based demographic signals were not observable, thereby anonymizing candidates with respect to *name-based* race/ethnicity and gender cues. (See Figures C.4 and C.6 for examples of the candidate-evaluation survey pages for HR and engineers under blind conditions.)

For participants in the Non-blind condition, each job applicant was assigned a hypothetical name drawn from a pool of common first and last names that signal three race/ethnicity groups (White, Asian, and Hispanic) and two genders (male and female). This allows us to vary applicants' implied demographics across six groups[11] while holding resume content constant, enabling estimation of group-specific discrimination and the treatment effect of blinding demographic signals. See Section C.II.2 for further discussion of name selection and the random assignment of demographic cues.

This design allows us to estimate how access to demographic signals affects evaluations at each stage of the hiring funnel, as well as across stages by matching HR and engineer evaluations. Because engineers in our experiment evaluated the full set of 18 candidates (rather than only the subset that HR would have shortlisted), interpreting these results as representing the full funnel requires an IIA-type assumption: engineers' relative rankings are not materially affected by the size or composition of the evaluated set, including candidates who would otherwise have been screened out by HR. We assess the plausibility of this assumption and report sensitivity analyses.

Incentives and IRR-style matching. Employer participants received a completion payment ($25 for HR professionals and $40 for software engineers). They were also entered into a performance-contingent lottery (six $50 prizes), with winning probabilities increasing in the accuracy of their predictions of applicants' coding ability, as measured by the coding-ability index described in Section II.A. In addition, participants received $10 for each successfully invited colleague who completed the survey. Finally, following the Incentivized Resume Rating (IRR) framework (Kessler et al., 2019), participants were asked whether they would like to receive resumes, coding submissions, and contact information for a subset of students whose profiles best matched the preferences they revealed in the task.[12]

II.C Outcome Variables

Our primary outcome measured on the job-applicant side (via the student coding lab) is an overall coding-score index. The submissions were graded by two independent software engineers with substantial industry experience, who evaluated only anonymized code. Grading followed a pre-specified rubric with the following components:

- **Accuracy (25%):** whether the code compiled passes pre-specified test cases.

[11] White male, White female, Asian male, Asian female, Hispanic male, and Hispanic female

[12] To limit potential discriminatory consequences for students, we do not incorporate protected-class variables such as race or gender into the matching procedure used to determine which student profiles shared.

- **Readability (25%):** subjective assessment by the two graders (12.5% each).
- **Theoretical efficiency (25%):** time and space complexity of the algorithm implemented in the code (i.e., theoretical computational steps and memory requirements).
- **Practical efficiency (25%):** relative runtime and compile time within each programming language (relative to other submissions in the same language).

The primary outcomes measured in the employer-side experiment/survey are:

1. **Perceived coding ability:** "How good do you think this person will be in coding?" (0–10).
2. **Workplace fit:** "How good of a fit do you think this person will be in your workplace?" (0–10).
3. **Retention:** "How likely would this person stay in your company for the next 5 years?" (0–10).
4. **Overall rating:** "How would you rate this person overall?" (0–100).
5. **Shortlist decision (HR only):** "Would you select this person to advance to the next hiring stage for more in-depth technical evaluation?" (Yes/No).

III Attrition, Balance, and Descriptive Statistics

III.A Attrition.

We test for differential attrition by comparing survey-completion rates across the Blind and Non-blind conditions, conditional on reaching the randomization stage and passing the quality check. As shown in Table A.2, we find no meaningful differential attrition between the experimental arms.

III.B Balance.

We assess randomization balance by regressing each pre-determined respondent and firm characteristic on the Blind-treatment indicator separately by evaluator role. Table A.3 shows that estimated differences are generally small and largely statistically indistinguishable from zero. While a couple coefficients reach conventional significance levels, the overall pattern is consistent with successful randomization of treatment of interest.

III.C Descriptive Statistics: Demographics.

Table A.1 summarizes respondent and firm characteristics for the employer sample ($N = 320$; 150 HR professionals and 170 software engineers). HR respondents are older on average than engineers (mean age 38.2 vs. 33.9) and are less likely to be male (0.33 vs. 0.81). Racial composition also differs by role: HR respondents are more likely to be White (0.57 vs. 0.45), Hispanic (0.10 vs.

0.05) or Black (0.05 vs. 0.01), and less likely to be Asian (0.25 vs. 0.44). Most of the respondents reported having a college degree (0.94 combined), and engineers are more likely to be in the highest income category ($\geq$ \$150K: 0.52 vs. 0.37). HR respondents report more hiring experience (9.2 vs. 5.8 years). On firm characteristics, most of the respondents report that their firm is currently hiring (0.79 overall), with representation across small, medium, and large firms.

III.D Descriptive Statistics: Attitudes on Diversity.

Figure 1 reports attitudes toward diversity (control group only). Overall, both HR and engineers express generally pro-diversity views, but HR respondents are consistently more pro-diversity than engineers: they rate the importance of working with a diverse workforce higher (Panel B: 4.08 vs. 3.60) and are more likely to view focusing on diversity as a good thing (Panel C: racial diversity 4.12 vs. 3.77; gender diversity 4.20 vs. 3.87). Both groups report similar perceived alignment between HR and engineering priorities on diversity (Panel A: 3.75 vs. 3.73). The perceived firm attention to diversity is moderate and higher among HR than among engineers (Panel D: racial 3.08 vs. 2.67; sex 3.10 vs. 2.80).

Optional open-ended responses are consistent with these patterns: HR respondents more often describe Diversity, equity, and inclusion (DEI) as an important hiring objective and emphasize structured processes and pipeline-building, while engineers more often stress technical merit and report limited formal DEI influence.

IV Results

IV.A Do Decision Makers at Different Hiring Stages Apply Different Hiring Rules?

We study our first research question: do decision-makers at different stages of hiring apply different decision rules based on the demographic identities of the candidates? Specifically, we examine whether HR professionals and software engineers assign different ratings to otherwise identical applicant profiles when demographic cues are conveyed through applicant names. Restricting to the *Non-blind* sample, we estimate the following specification separately by evaluator role:

$$y_{i,s} = \alpha_0 + \beta_1 Race_Gender_{i,s} + \mu_i + \omega_{i,s} + \delta_s + \varepsilon_{i,s}, \tag{1}$$

where $y_{i,s}$ is respondent s's standardized evaluation of resume i for the outcomes listed in Section II.C (employer phase). *Race_Gender*$_{i,s}$ is a set of indicator variables for the applicant demographic group implied by the assigned name (White/Asian/Hispanic $\times$ male/female), with *White male* as the omitted category. μ_i denotes resume fixed effects (holding constant the underlying resume content), $\omega_{i,s}$ denotes fixed effects for the resume's position/order in respondent s's evaluation sequence, and δ_s denotes respondent fixed effects. $\varepsilon_{i,s}$ are the standard errors clustered at the respondent level.

Table A.4 columns (1)–(5) and (6)–(10) report the estimated coefficients for HR and engineers, respectively. Each coefficient can be interpreted as the premium (if positive) or penalty (if negative) associated with assigning a given demographic cue (via name) to the *same* resume profile, relative to assigning a White male name.

In the HR sample (Panel A), we find little evidence on differential ratings by demographics. The main exception is that resumes assigned Hispanic male names receive higher scores on coding ability and overall rating than the same resumes assigned White male names: 0.153 standard deviations (SD) higher predicted coding ability and 0.157 SD higher overall ratings (and 0.156 SD higher on the composite z-score).

In the engineer sample (Panel B), several non-White name groups receive positive premia relative to White male names. In particular, Asian male names are rated higher on predicted coding ability (0.131 SD) and overall ratings (0.082 SD), Asian female names receive higher workplace fit (0.112 SD) and overall ratings (0.094 SD), Hispanic male names receive higher predicted coding ability (0.070 SD) and overall ratings (0.069 SD), and Hispanic female names receive higher predicted coding ability (0.112 SD), workplace fit (0.101 SD), and overall ratings (0.132 SD).

Across outcomes, differences between HR and engineer coefficients are generally not statistically distinguishable (Panel C), though we find suggestive evidence of larger engineer premia for Asian male predicted coding ability and Hispanic female overall ratings (both $p = 0.092$).

Notably, these name-based effects appear even in the technical evaluation stage, where engineers observe more information about candidates, such as coding submissions and objective performance information. This suggests that name-based demographic cues can still affect evaluations, even when evaluators observe detailed applicant information with more direct or precise signals of productivity.

IV.B How Does Blinding Affect Employers' Candidate Evaluations?

We next turn to study our second question: how does blinding the observable group characteristics affect employers' subjective evaluation of job applicants? To do so, we estimate the causal effect of blinding applicant names (i.e., removing name-based demographic cues) on employers' evaluations, and how this effect varies with the race/ethnicity and gender implied by the name assigned to the resumes. We estimate the treatment effects separately for HR professionals and software engineers.

Treatment effects by implied race/gender. For each evaluator role, we estimate:

$$y_{i,s} = \gamma_0 - \sum_{d \in \mathcal{G}} \gamma_d \, \mathbf{1}\{Race_Gender_{i,s} = d\} + \mu_i + \omega_{p(i,s)} + X_s'\beta + \varepsilon_{i,s}, \tag{2}$$

where $y_{i,s}$ is respondent s's standardized evaluation of profile i (Section II.C). $Race_Gender_{i,s}$ denotes the name-implied demographics shown to respondent s for profile i and takes values in $\{\textit{Blind}\} \cup \mathcal{G}$, where $\mathcal{G} = \{White, Asian, Hispanic\} \times \{Male, Female\}$. Note that we treat the blinded

resumes as an independent demographic category for this analysis, unlike in equation 1, and treat this category as an omitted base group. μ_i denotes resume fixed effects (holding constant underlying resume content), $\omega_{p(i,s)}$ denotes fixed effects for the profile's position/order in respondent s's evaluation sequence, and X_s includes pre-determined respondent controls (e.g., demographics and background characteristics). Standard errors are clustered at the respondent level.

Because *Blind* is the omitted category and we parameterize the regression with a leading minus sign, each coefficient γ_d represents the treatment effect of blinding a specific demographic group ($d \in \mathcal{G}$):

$$\gamma_d = \mathbb{E}[y_{i,s} \mid Race_Gender_{i,s} = \textit{Blind}] - \mathbb{E}[y_{i,s} \mid Race_Gender_{i,s} = d],$$

Therefore, a positive γ_d is interpreted as the premium (or benefit) that demographic group d receives when names are blinded to employers, while a negative γ_d is interpreted as the penalty that demographic group d incurs when their names are blinded.

Table 2 reports the implied group-specific treatment effects of blinding (Blind minus Non-blind) for each demographic group and outcome, separately by role. For HR (Panel A), blinding substantially lowers evaluations for resumes assigned Hispanic male names, including predicted coding ability and overall ratings (as well as the composite z-score). In contrast, the estimated effects for other groups are comparatively small, with limited evidence of systematic shifts. For engineers (Panel B), blinding increases ratings for White male candidates on several outcomes, while reducing ratings for groups that received positive name-based premia in the Non-blind condition (notably Asian male and Hispanic female candidates on key dimensions).

The treatment effects are consistent with the name-based premia/penalties documented in Section IV.A. In particular, groups that receive positive premia when names are visible tend to experience negative treatment effects when names are removed (and vice versa). This pattern is mechanically consistent with blinding operating through the removal of demographic cues, rather than changes in resume content.

IV.C How does Blind Hiring Interact with Multiple Hiring Stages?

We study our last research question: how does blind hiring interact with multiple stages of hiring and alter the composition or productivity of candidates who are ultimately hired? To answer this question, we ex-post construct matching pairs between HR and engineer respondents and two-stage treatment arms, described below.

Creating matched HR-Engineer pairs. Our experiment produces separate evaluations of the same standardized applicant profiles by HR professionals and software engineers. To study how blinding propagates through a two-stage hiring funnel, we construct synthetic hiring "pairs" by matching an HR professional h to an engineer e whenever both evaluated the exact same appli-

cation version (same resume content and the same assigned name).[13,14] Let $\mathcal{J} = (h, e)$ denote an HR–engineer pair.

Two-Stage Treatment Arms. Each pair $\mathcal{J}$ belongs to one of three pair-level information conditions depending on whether the demographic cues conveyed by the names are visible at each stage:

1. *Control* (*Non-blind*)*:* both h and e observe non-blinded applications;
2. *Realistic blind hiring:* h observes blinded applications while e observes non-blinded applications;
3. *Complete blind hiring:* both h and e observe blinded applications.

Mapping evaluations to hiring outcomes. To define the "hired" applicant for each pair $\mathcal{J}$, we map the HR-stage and engineer-stage evaluations into a two-stage selection rule. Let $\mathcal{C}$ denote the number of applicants advanced from HR to the engineer stage. We implement two approaches for defining the HR-stage advancement pool.

First, we impose an *exogenous* HR-stage selection rule that holds the screening intensity fixed: for a given $\mathcal{C}$, we rank the 18 applicants reviewed by h using HR's overall evaluation score and advance the top $\mathcal{C}$ applicants.[15] For now, we consider $\mathcal{C} \in \{5, 10\}$, which correspond to passing a relatively selective versus relatively inclusive set of candidates to the technical stage.

Second, we use an *endogenous* HR-stage selection rule that mirrors HR discretion: we advance the set of applicants that h explicitly marked as *Advance* (a binary 0/1 decision), allowing $\mathcal{C}$ to vary across HR professionals and potentially across treatment conditions. Importantly, this binary advancement decision is not mechanically tied to HR's continuous overall rating, though very highly correlated: in practice, an applicant with a higher overall score may be marked *Do not advance* while an applicant with a slightly lower overall score is marked *Advance*. Accordingly, the endogenous rule captures a discretionary screening margin that is distinct from a simple rank-order cutoff based on overall scores.

Given the HR-stage advancement set for pair $\mathcal{J}$, the engineer then selects among the advanced candidates based on the engineer's overall evaluation score. Specifically, for $K \in \{1, 2, 3\}$, we define the engineer's selection as the top K candidates ranked by e within the HR-advanced pool. Let $i^{\mathcal{J}}$ denote the candidate selected by pair $\mathcal{J}$ when $K = 1$ (and analogously define the set of selected candidates for $K > 1$).

How blinding affects the demographics of selected candidates. To see the impact of blinding (both HR and Engineers) on the probability of hiring for each demographic groups, we restrict

[13]Matching is described in more detail in Section C.II.2.

[14]This implies that each HR or engineer will appear in more than one pair in the pair-level regression. Inference accounts for this by clustering standard errors at the HR and engineer levels.

[15]When ties occur at the cutoff, we break ties using predicted coding score and then random selection.

attention to the *Complete BH* and *Control* arms and estimate, for each choice of HR-pool rule and engineer selection rule:

$$y_{i,\mathcal{J}} = \gamma_0 - \sum_{d \in \mathcal{G}} \gamma_d \, \mathbf{1}\{Race_Gender_{i\mathcal{J}} = d\} \; + \; X'_{\mathcal{J}}\lambda \; + \; \mu_i \; + \; \delta_{order} \; + \; \varepsilon_{i,\mathcal{J}}, \tag{3}$$

where $y_{i,\mathcal{J}}$ is a binary variable equal to 1 if resume i is selected, or 0 otherwise; $Race_Gender_{i\mathcal{J}}$ denotes the demographic groups implied by name shown to pair $\mathcal{J}$ for resume i and takes values in $\{$*Complete Blind Hiring*$\} \cup \mathcal{G}$, where $\mathcal{G} = \{White, Asian, Hispanic\} \times \{Male, Female\}$.

Tables 3 and 4 report how *Complete* blind hiring changes selection probabilities for each race–gender group when the outcome is an indicator for whether a given applicant is selected (hired) under the two-stage rule described above.[16] Table 3 holds the HR-stage advancement rule fixed (*exogenous* HR pool) to advancing the top 5 and 10 candidates, while Table 4 uses HR's own *Advance* decisions (*endogenous* HR pool).

Two patterns emerge. First, when the HR-stage pool is relatively selective (HR passes Top 5), treatment effects are small and statistically indistinguishable from zero across groups regardless of whether engineers select Top 1, Top 2, or Top 3 candidates. Second, when the HR-stage pool is more inclusive (HR passes Top 10), Complete blind hiring shifts selection probabilities in a direction that becomes more pronounced as the engineer selects more finalists: the probability that a white male applicant is selected increases ($+0.028$ and $+0.047$ when Engineer hires its top 2 and 3 choices among those passed by the HR), while the probability that an Asian female applicant is selected decreases (-0.034 and -0.043 for $k \in \{2, 3\}$). Under the endogenous HR pool (Table 4), the similar pattern appears: Complete blind hiring increases selection of white male applicants when engineers select Top 2 ($+0.023$) or Top 3 ($+0.036$), and it increases selection of Hispanic female applicants when engineers select only Top 1 ($+0.021$), with other group-level effects comparatively small.

How blinding shifts the productivity of selected candidates. Next, we examine whether blinding affects not only *who* is hired, but also the *quality* of the selected candidate as measured by our pre-determined coding-ability score. For each HR–engineer pair $\mathcal{J}$ and each definition of the HR-stage advancement pool—either an exogenous rule that advances the Top $\mathcal{C}$ applicants (e.g., $\mathcal{C} \in \{5, 10\}$) or the endogenous set of applicants marked *Advance* by HR—we identify the candidate ultimately selected by the engineer. Then, using the measure of candidate's productivity using the coding score from the proctored lab assessment, we study how blinding changes the coding ability of selected candidates. In particular, we estimate

we estimate

$$y_{\mathcal{J}} = \gamma_0 + \gamma_1 Realistic_{\mathcal{J}} + \gamma_2 Complete_{\mathcal{J}} + \varepsilon_{\mathcal{J}}, \tag{4}$$

where $y_{\mathcal{J}}$ is the actual coding score of the selected candidate measured via the student coding lab.

[16] Positive coefficients indicate that blinding increases the probability of being selected; negative coefficients indicate that blinding decreases selection. Standard errors are two-way clustered by HR and engineer.

Tables 5-6 show that blinding at the HR stage alone (Realistic BH) has little effect on the coding ability of selected candidates: estimated effects are close to zero and generally statistically insignificant (with at most a modest decline when HR passes Top 5 and the engineer selects Top 3). In contrast, Complete blind hiring (blinding at both the HR and engineer stages) consistently reduces the coding score of the selected candidate. Under the endogenous HR advancement rule (Table 5), Complete BH similarly reduces selected-candidate coding ability, with declines on the order of 0.04–0.05 standard deviations when the engineer selects Top 1 or Top 2 candidates (and a smaller, statistically weaker decline when selecting Top 3). Under the exogenous HR advancement rules (Table 6), Complete BH lowers the hired candidate's coding score by roughly 0.04–0.07 standard deviations across both $\mathcal{C} = 5$ and $\mathcal{C} = 10$ and across alternative engineer-selection rules.

We are cautious in interpreting these productivity declines as evidence that name-based demographic cues convey useful information about applicant quality and thereby blinding allows useful information to be taken away. In our design, names are experimentally assigned, and the same underlying candidate resume is paired with all possible race–gender identity signals; as a result, names do not contain genuine, applicant-specific information about coding ability. Blinding therefore removes a signal that is, by construction, orthogonal to true productivity and may be noisy or even misleading. More work is needed to understand the results.

IV.D Heterogeneity Effects of Blind Hiring

Work is in progress to examine the heterogeneous treatment effects of blind hiring.

V Conclusion

TBD

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Main Tables

Table 1: Treatment Effects of Blind Hiring By Employers' Role

	(1) Std:Overall
Panel A: HR	
White Male	0.037
	(0.057)
	[0.935]
White Female	0.013
	(0.051)
	[0.961]
Asian Male	0.014
	(0.059)
	[0.961]
Asian Female	-0.023
	(0.048)
	[0.954]
Hispanic Male	-0.115**
	(0.048)
	[0.094]
Hispanic Female	0.086*
	(0.052)
	[0.380]
Blind Group Mean	0.037
N	2466
Panel B: Engineer	
White Male	0.086***
	(0.028)
	[0.025]
White Female	0.023
	(0.035)
	[0.849]
Asian Male	-0.017
	(0.028)
	[0.849]
Asian Female	-0.029
	(0.029)
	[0.747]
Hispanic Male	-0.003
	(0.030)
	[0.924]
Hispanic Female	-0.068**
	(0.034)
	[0.179]
Blind Group Mean	0.086
N	2772
p-val. HR-Eng: White Male	0.441
p-val. HR-Eng: White Female	0.865
p-val. HR-Eng: Asian Male	0.626
p-val. HR-Eng: Asian Female	0.910
p-val. HR-Eng: Hispanic Male	0.047
p-val. HR-Eng: Hispanic Female	0.012

Notes: Base group is treatment group. Resume fixed effects, resume order fixed effects, and respondent controls (i.e., race, gender, income, education) are included. The Westfall-Young p-values adjusted for multiple hypothesis testing are in brackets. Any adjusted p-values greater than 1 is manually set to be equal to 1 according to the Westfall-Young p-value adjustment,

Table 2: Treatment Effects of Blind Hiring By Employers' Role

	(1) Std:Coding	(2) Std:Fit	(3) Std:Stay	(4) Advance
	Panel A: HR			
White Male	0.038	-0.051	-0.071	0.031
	(0.056)	(0.058)	(0.058)	(0.049)
	[0.999]	[0.998]	[0.965]	[0.999]
White Female	0.019	-0.004	-0.027	0.023
	(0.046)	(0.054)	(0.063)	(0.048)
	[0.999]	[0.999]	[0.999]	[0.999]
Asian Male	0.032	-0.014	0.043	0.047
	(0.052)	(0.062)	(0.060)	(0.047)
	[0.999]	[0.999]	[0.999]	[0.993]
Asian Female	0.020	0.027	-0.026	0.044
	(0.052)	(0.057)	(0.060)	(0.050)
	[0.999]	[0.999]	[0.999]	[0.998]
Hispanic Male	-0.142***	0.068	0.091	0.073
	(0.049)	(0.054)	(0.061)	(0.046)
	[0.117]	[0.965]	[0.893]	[0.849]
Hispanic Female	0.040	0.057	-0.011	0.028
	(0.058)	(0.067)	(0.073)	(0.055)
	[0.999]	[0.998]	[0.999]	[0.999]
Blind Group Mean	0.038	-0.051	-0.071	0.031
N	2466	2466	2466	2466
	Panel B: Engineer			
White Male	0.072**	-0.040	-0.045	
	(0.029)	(0.041)	(0.049)	
	[0.215]	[0.943]	[0.954]	
White Female	0.028	-0.046	-0.023	
	(0.031)	(0.045)	(0.052)	
	[0.954]	[0.941]	[0.990]	
Asian Male	-0.071**	-0.083*	-0.002	
	(0.028)	(0.046)	(0.049)	
	[0.193]	[0.565]	[0.992]	
Asian Female	0.003	0.099**	0.053	
	(0.029)	(0.042)	(0.048)	
	[0.992]	[0.233]	[0.941]	
Hispanic Male	0.009	-0.023	-0.043	
	(0.027)	(0.037)	(0.054)	
	[0.990]	[0.973]	[0.954]	
Hispanic Female	-0.050*	0.101**	0.067	
	(0.026)	(0.042)	(0.058)	
	[0.501]	[0.217]	[0.941]	
Blind Group Mean	0.072	-0.040	-0.045	
N	2772	2772	2772	
p-val. HR-Eng: White Male	0.586	0.875	0.731	
p-val. HR-Eng: White Female	0.871	0.545	0.955	
p-val. HR-Eng: Asian Male	0.077	0.369	0.558	
p-val. HR-Eng: Asian Female	0.771	0.309	0.295	
p-val. HR-Eng: Hispanic Male	0.007	0.161	0.096	
p-val. HR-Eng: Hispanic Female	0.152	0.578	0.393	

Notes: Base group is treatment group. Resume fixed effects, resume order fixed effects, and respondent controls (i.e., race, gender, income, education) are included. The Westfall-Young p-values adjusted for multiple hypothesis testing are in brackets. Any adjusted p-values greater than 1 is manually set to be equal to 1 according to the Westfall-Young p-value adjustment.

Table 3: Treatment Effects on Selection Probability: Exogenous HR Selection Rule

	Dependent Variable: Binary Variable for Hired					
	HR passes Top 5 Candidates			**HR passes Top 10 Candidates**		
	Engineer Hires Top 1 (1)	Engineer Hires Top 2 (2)	Engineer Hires Top 3 (3)	Engineer Hires Top 1 (4)	Engineer Hires Top 2 (5)	Engineer Hires Top 3 (6)
White Male	0.004	0.004	0.007	0.012	0.028**	0.047***
	(0.009)	(0.014)	(0.019)	(0.009)	(0.013)	(0.015)
White Female	-0.001	0.006	0.008	-0.003	-0.010	-0.006
	(0.011)	(0.015)	(0.020)	(0.013)	(0.015)	(0.016)
Asian Male	-0.009	-0.005	-0.0005	-0.012	-0.013	-0.010
	(0.009)	(0.012)	(0.016)	(0.011)	(0.014)	(0.014)
Asian Female	-0.001	-0.004	-0.004	-0.015	-0.034**	-0.043***
	(0.008)	(0.014)	(0.019)	(0.011)	(0.013)	(0.015)
Hispanic Male	0.005	0.001	-0.010	0.004	0.015	0.006
	(0.008)	(0.012)	(0.017)	(0.008)	(0.011)	(0.013)
Hispanic Female	0.002	-0.002	0.0006	0.016**	0.015	0.006
	(0.008)	(0.013)	(0.019)	(0.007)	(0.011)	(0.014)
Respondent Controls	✓	✓	✓	✓	✓	✓
Resume FE	✓	✓	✓	✓	✓	✓
Display Order FE	✓	✓	✓	✓	✓	✓
Observations	136,242	136,242	136,242	136,242	136,242	136,242
Adjusted R^2	0.13091	0.19385	0.21117	0.15446	0.24377	0.29686

Standard errors two-way clustered by HR and Engineer.
Comparison: Complete BH vs Control.
Positive (+) = Blinding INCREASED hiring. Negative (-) = Blinding DECREASED hiring.

Table 4: Treatment Effect of Blind Hiring on Selection Probability (Endogenous HR Pool)

	Dependent Variable: Binary Variable for Hired		
	Engineer Select Conditional on HR Advancement		
	Engineer Hires Top 1 (1)	Engineer Hires Top 2 (2)	Engineer Hires Top 3 (3)
White Male	0.008	0.023*	0.036**
	(0.010)	(0.013)	(0.016)
White Female	-0.010	-0.015	-0.007
	(0.013)	(0.016)	(0.019)
Asian Male	-0.009	-0.008	-0.009
	(0.011)	(0.015)	(0.019)
Asian Female	-0.002	-0.008	-0.012
	(0.011)	(0.014)	(0.018)
Hispanic Male	0.004	0.011	0.011
	(0.008)	(0.011)	(0.014)
Hispanic Female	0.021***	0.017	0.012
	(0.007)	(0.012)	(0.015)
Respondent Controls	✓	✓	✓
Resume FE	✓	✓	✓
Display Order FE	✓	✓	✓
Observations	136,242	136,242	136,242
Adjusted R^2	0.13630	0.21174	0.25895

Standard errors two-way clustered by HR and Engineer.
Comparison: Complete BH vs Control (Nonblind).
Sample: Candidates advanced by HR (Endogenous).
Positive (+) = Blinding INCREASED hiring.
Negative (-) = Blinding DECREASED hiring.

Table 5: Treatment Effects on Productivity (Coding Ability) of Hired: Exogenous HR Selection Rule

	Dependent Variable: Coding Score (Z)					
	HR passes Top 5 Candidates			**HR passes Top 10 Candidates**		
	Engineer Hires Top 1 (1)	Engineer Hires Top 2 (2)	Engineer Hires Top 3 (3)	Engineer Hires Top 1 (4)	Engineer Hires Top 2 (5)	Engineer Hires Top 3 (6)
Realistic BH (Blind HR)	-0.003	-0.010	-0.040*	-0.003	-0.001	-0.0003
	(0.013)	(0.011)	(0.021)	(0.007)	(0.009)	(0.007)
Complete BH (Blind HR + Eng)	-0.050*	-0.048**	-0.069***	-0.044**	-0.063**	-0.047**
	(0.026)	(0.021)	(0.024)	(0.021)	(0.028)	(0.024)
Resume FE	✓	✓	✓	✓	✓	✓
Display Order FE	✓	✓	✓	✓	✓	✓
Control Arm Mean	0.000	0.000	0.000	0.000	0.000	0.000
Observations	14,121	28,242	42,363	14,121	28,242	42,363
Adjusted R^2	0.90124	0.89391	0.87554	0.93219	0.89545	0.89065

Standard errors two-way clustered by HR and Engineer.
Columns 1-3: Engineer selects from HR's Top 5.
Columns 4-6: Engineer selects from HR's Top 10.

Table 6: Treatment Effects on Productivity (Coding Ability) of Hired: Endogenous HR Selection Rule

	Engineer Select from Candidates Endogenously Advanced by HR		
	overall_z_dyn		
	Hires Top 1	Hires Top 2	Hires Top 3
	(1)	(2)	(3)
Realistic BH (Blind HR)	-0.003	-0.001	0.006
	(0.009)	(0.016)	(0.023)
Complete BH (Blind HR + Eng)	-0.039*	-0.054**	-0.033
	(0.021)	(0.025)	(0.027)
Resume Controls	✓	✓	✓
Display Order FE	✓	✓	✓
Control Arm Mean	0.000	0.000	0.000
Observations	13,744	27,135	40,347
Adjusted R^2	0.95136	0.91636	0.89708

Standard errors two-way clustered by HR and Engineer.

Sample: Candidates actually advanced by HR.

Main Figures

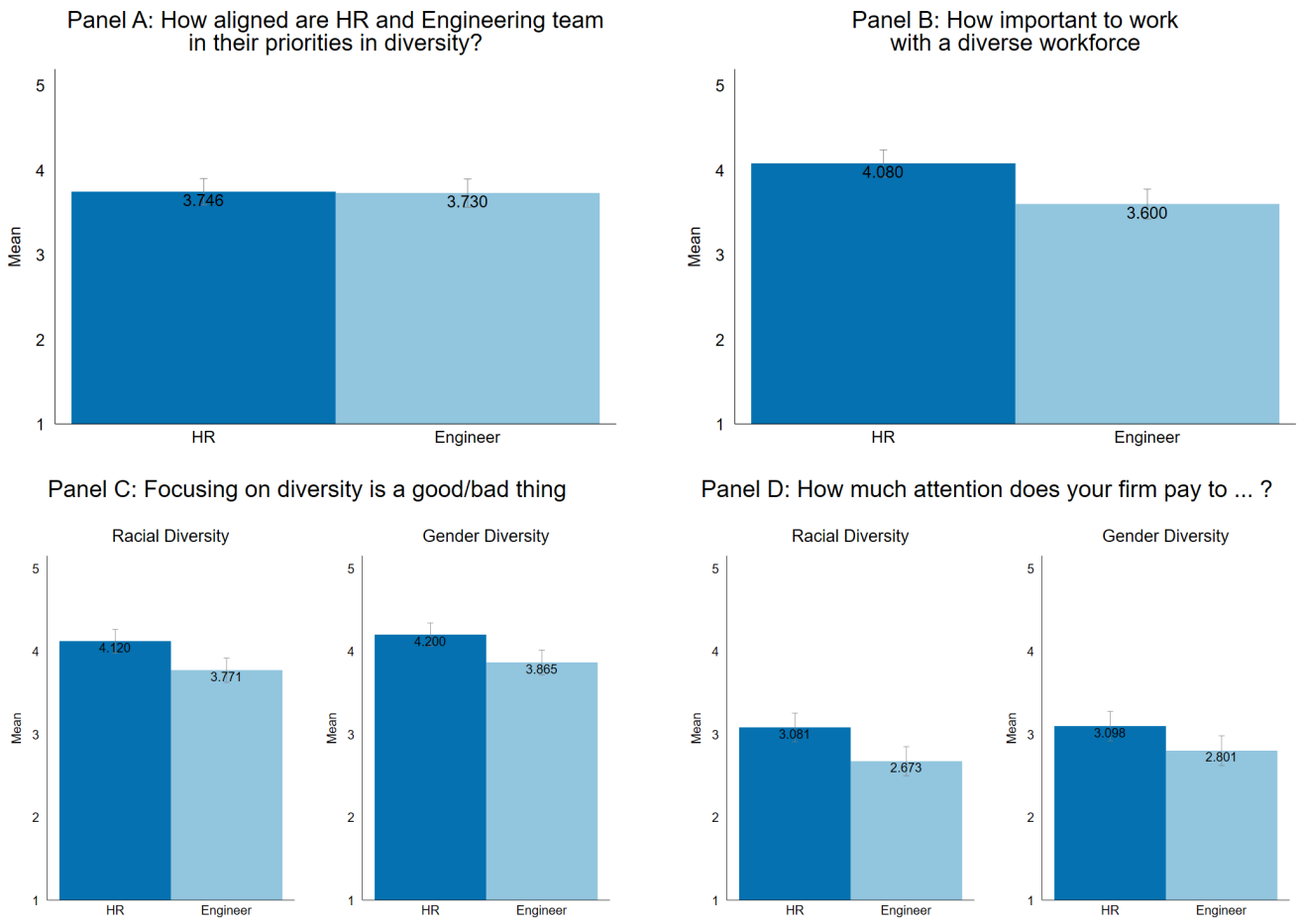
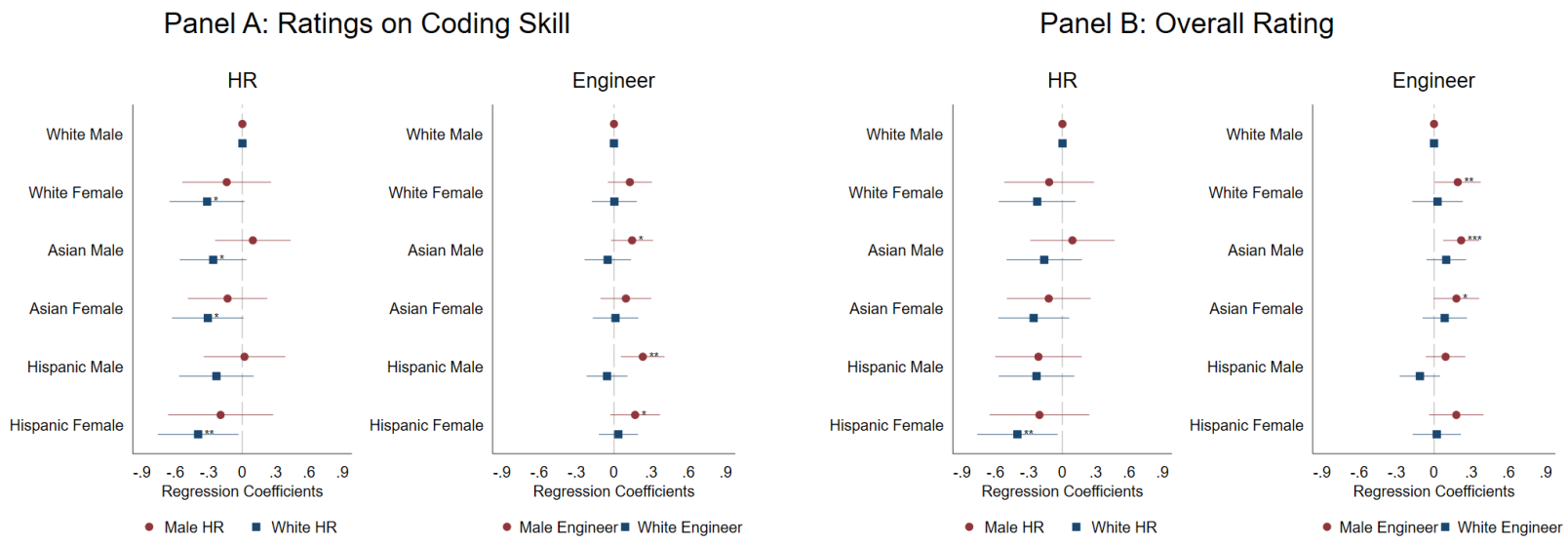
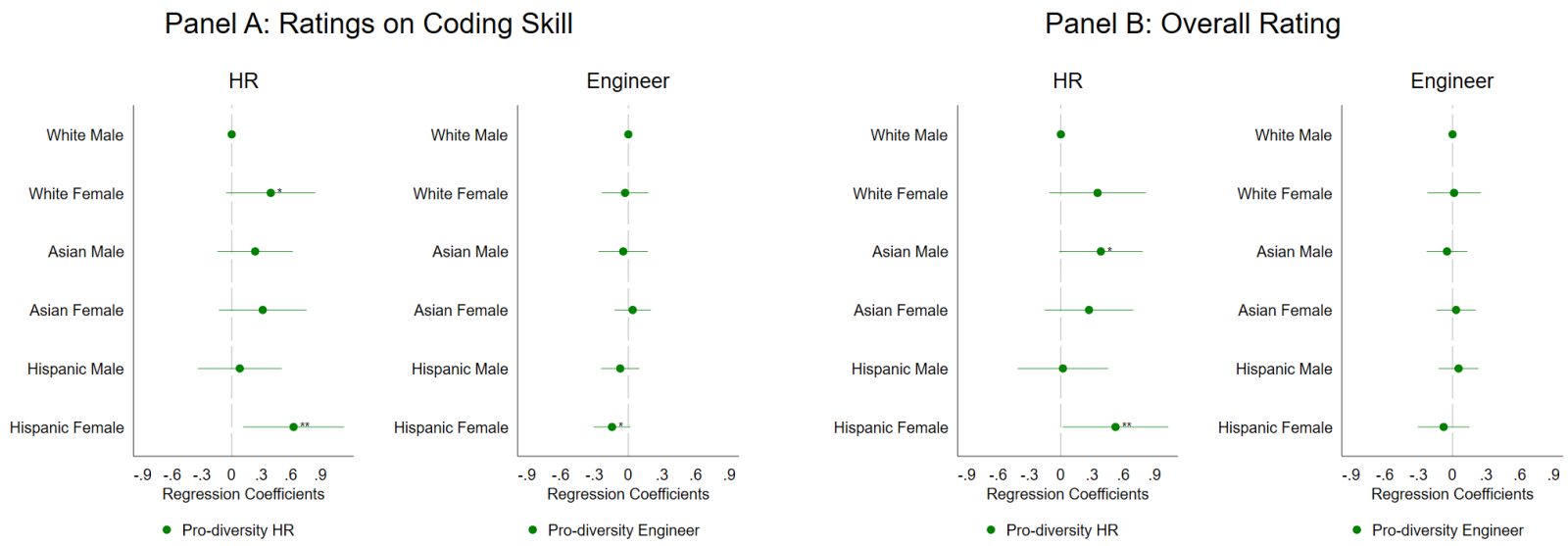


Figure 1: Views on Diversity by HRs and Engineers
(Control Group Only)



Notes: Panels A and B control for race for male hetero; control for gender for race hetero. Panels C and D control for race and gender.

Figure 2: Heterogeneity: Ratings by Employers' Role and Identity
(Non-blind Condition Only)



Notes: Panels A and B control for race for male hetero; control for gender for race hetero. Panels C and D control for race and gender.

Figure 3: Heterogeneity: Ratings by Employers' Role and Diversity Preference (Non-blind Condition Only)

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A Appendix Tables

Table A.1: Employer Survey: Summary Statistics

	Scale	All		HR		Engineers	
		Mean (SD)	N	Mean (SD)	N	Mean (SD)	N
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Respondent Characteristics							
Age	C	35.88 (9.48)	320	38.17 (9.35)	150	33.87 (9.16)	170
Gender: Male	B	0.58 (0.49)	320	0.33 (0.47)	150	0.81 (0.39)	170
Race: White	B	0.51 (0.50)	320	0.57 (0.50)	150	0.45 (0.50)	170
Race: Asian	B	0.35 (0.48)	320	0.25 (0.44)	150	0.44 (0.50)	170
Race: Hispanic	B	0.09 (0.29)	320	0.10 (0.30)	150	0.08 (0.28)	170
Race: Black	B	0.03 (0.17)	320	0.05 (0.23)	150	0.01 (0.11)	170
Race: Other	B	0.02 (0.14)	320	0.03 (0.16)	150	0.01 (0.11)	170
College Degree	B	0.94 (0.23)	320	0.92 (0.27)	150	0.96 (0.19)	170
Annual Income $<$ 100K USD	B	0.26 (0.44)	320	0.33 (0.47)	150	0.19 (0.39)	170
100 K $\geq$ Annual Income $<$ 150K USD	B	0.29 (0.46)	320	0.30 (0.46)	150	0.29 (0.45)	170
150 K $\geq$ Annual Income	B	0.45 (0.50)	320	0.37 (0.48)	150	0.52 (0.50)	170
At Current Firm (in years)	C	10.04 (112.76)	320	3.64 (3.47)	150	15.68 (154.67)	170
Hiring Experience (in years)	C	7.40 (6.80)	320	9.22 (6.70)	150	5.79 (6.50)	170
Panel B: Firm Chracteristics							
Currently Hiring	B	0.79 (0.41)	320	0.91 (0.29)	150	0.69 (0.46)	170
Firm Size: Small	B	0.37 (0.48)	320	0.25 (0.44)	150	0.48 (0.50)	170
Firm Size: Medium	B	0.38 (0.49)	320	0.46 (0.50)	150	0.32 (0.47)	170
Firm Size: Large	B	0.24 (0.43)	320	0.29 (0.45)	150	0.21 (0.41)	170

Notes: C refers to continuous, B refers to binary.

Table A.2: Testing for differential attrition

	Outcome: Completed Survey		
	(1)	(2)	(3)
	All	HR	Enigneer
Treatment: Blind	0.042	0.018	0.061
	(0.040)	(0.058)	(0.055)
Control Mean	0.042	0.018	0.061
N	500	228	272

Notes: Table tests differential attrition between the control and blind treatment group from the sample of the employer survey. The sample includes participants who reached the randomization stage and passed the quality check. The outcome variable is a binary variable equal to 1 if the participant completed the survey or 0 otherwise. Robust standard errors are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.3: Treatment Balance by Employers' Role

	(1) Age	(2) Male	(3) Race: White	(4) Race: Asian	(5) Race: Hispanic	(6) Race: Black	(7) Race: Other	(8) College Degree	(9) Income Group 1	(10) Income Group 2	(11) Income Group 3	(12) At Current Firm	(13) Hiring Experience	(14) Firm is Hiring	(15) Small Firm	(16) Medium Firm	(17) Big Firm
Panel A: HR																	
Treatment: Blind	0.053	0.067	0.128	-0.047	-0.048	-0.031	-0.002	-0.020	-0.080	0.176**	-0.096	0.226	-0.372	0.088*	0.113	-0.104	-0.010
	(1.543)	(0.077)	(0.081)	(0.072)	(0.050)	(0.037)	(0.027)	(0.044)	(0.077)	(0.073)	(0.079)	(0.570)	(1.101)	(0.048)	(0.071)	(0.082)	(0.074)
Control Mean	0.053	0.067	0.128	-0.047	-0.048	-0.031	-0.002	-0.020	-0.080	0.176	-0.096	0.226	-0.372	0.088	0.113	-0.104	-0.010
N	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150
Panel B: Engineer																	
Treatment: Blind	-0.068	0.028	0.048	0.048	-0.073*	-0.024	-0.000	0.001	-0.051	0.005	0.047	-23.134	0.127	0.090	0.118	-0.149**	0.030
	(1.412)	(0.060)	(0.077)	(0.077)	(0.042)	(0.017)	(0.017)	(0.028)	(0.060)	(0.070)	(0.077)	(24.011)	(1.000)	(0.071)	(0.077)	(0.071)	(0.062)
Control Mean	-0.068	0.028	0.048	0.048	-0.073	-0.024	0.000	0.001	-0.051	0.005	0.047	-23.134	0.127	0.090	0.118	-0.149	0.030
N	170	170	170	170	170	170	170	170	170	170	170	170	170	170	170	170	170

Table A.4: Demographic-Based Discrimination in Applicant Ratings by Employers' Role
(Non-blind Condition Only)

	(1) Std:Coding	(2) Std:Fit	(3) Std:Stay	(4) Std:Overall	(5) z-score	(6) Std:Coding	(7) Std:Fit	(8) Std:Stay	(9) Std:Overall	(10) z-score	(11)	(12)	(13)	(14)	(15)
											p-val.: HR - Engineer				
	Panel A: HR					Panel B: Engineer					(1)-(6)	(2)-(7)	(3)-(8)	(4)-(9)	(5)-(10)
White Female	0.052	0.098	0.030	0.069	0.063	0.018	-0.029	0.005	0.026	0.007	0.726	0.251	0.251	0.659	0.605
	(0.089)	(0.090)	(0.099)	(0.084)	(0.092)	(0.046)	(0.067)	(0.077)	(0.050)	(0.059)					
Asian Male	-0.021	0.029	0.086	0.027	0.042	0.131**	0.002	0.073	0.082*	0.095*	0.092	0.797	0.797	0.556	0.577
	(0.077)	(0.084)	(0.092)	(0.085)	(0.083)	(0.050)	(0.066)	(0.067)	(0.042)	(0.052)					
Asian Female	0.050	0.077	0.011	0.076	0.051	0.063	0.112*	0.093	0.094**	0.093	0.890	0.719	0.719	0.839	0.657
	(0.082)	(0.076)	(0.093)	(0.076)	(0.079)	(0.045)	(0.060)	(0.074)	(0.044)	(0.057)					
Hispanic Male	0.153*	0.115	0.119	0.157*	0.156*	0.070*	-0.002	-0.010	0.069*	0.027	0.378	0.246	0.246	0.344	0.170
	(0.088)	(0.084)	(0.089)	(0.085)	(0.082)	(0.040)	(0.058)	(0.063)	(0.040)	(0.048)					
Hispanic Female	0.021	0.149	0.033	-0.042	0.045	0.112***	0.101*	0.106	0.132**	0.121**	0.364	0.676	0.676	0.092	0.517
	(0.093)	(0.100)	(0.111)	(0.092)	(0.102)	(0.042)	(0.058)	(0.082)	(0.051)	(0.060)					
Mean: White Male	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000					
N	1296	1296	1296	1296	1296	1512	1512	1512	1512	1512					

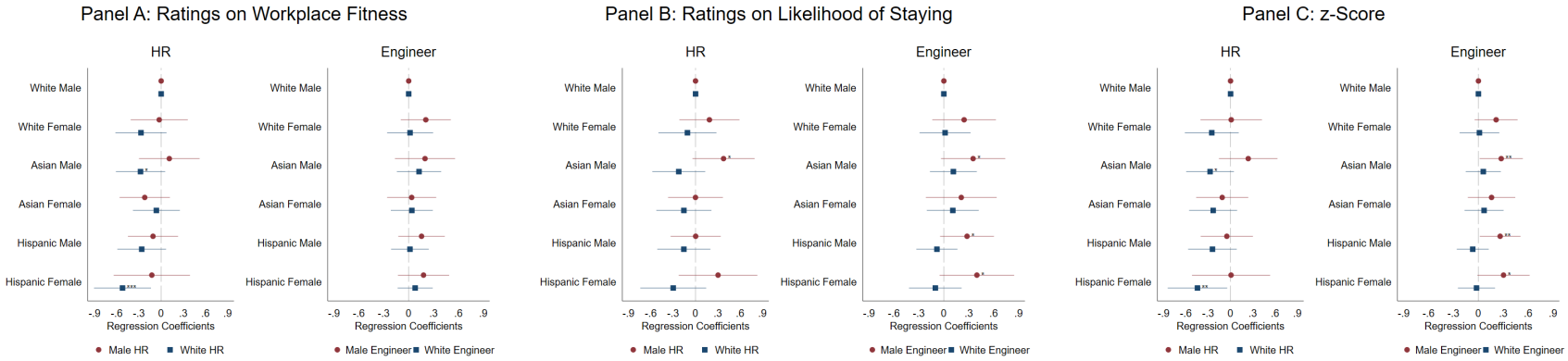
Notes: Base group is white male. Control sample only. Resume fixed effects, resume order fixed effects, and respondent FEs are included. SE clustered by respondent.

Table A.5: Treatment Effect of Blind Hiring with Resume Characteristics

	HR Std:Overall	Engineer Std:Overall
White Male	0.025	0.079***
	(0.057)	(0.027)
White Female	-0.006	0.031
	(0.058)	(0.034)
Asian Male	0.037	0.004
	(0.062)	(0.026)
Asian Female	-0.046	-0.039
	(0.049)	(0.029)
Hispanic Male	-0.066	-0.006
	(0.051)	(0.030)
Hispanic Female	0.064	-0.078**
	(0.050)	(0.032)
GPA	-0.537***	-0.171***
	(0.025)	(0.015)
Test Cases Passed		-0.763***
		(0.016)
Order FE	Yes	Yes
Respondent FE	No	No
Resume FE	No	No
R-squared	0.278	0.714
Observations	2628	2916

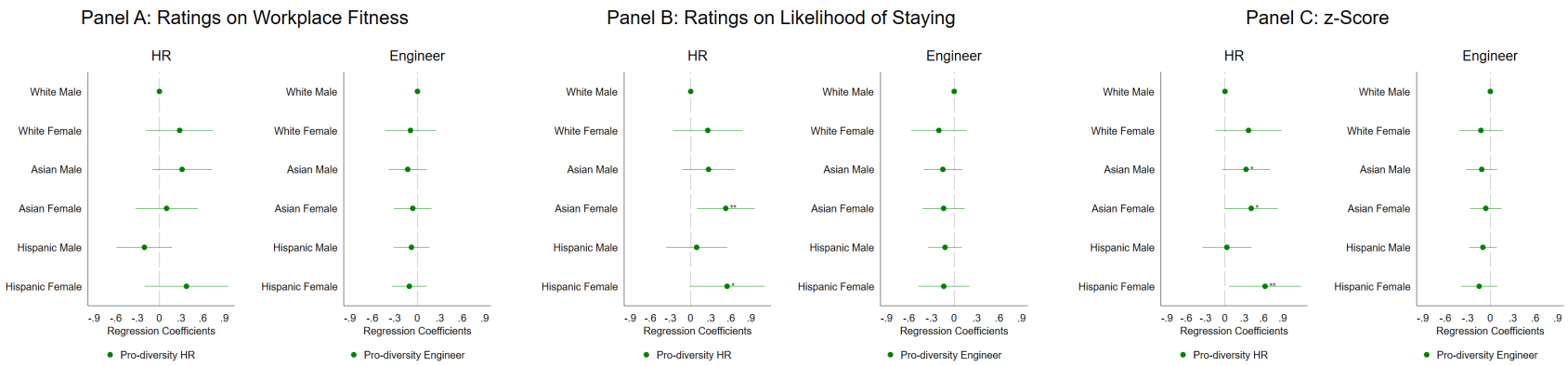
Notes: Included respondent characteristics like race, gender, income and education as controls.

B Appendix Figures



Notes:

Figure B.1: Heterogeneity: Ratings by Employers' Role and Identity - Additional Outcomes (Non-blind Condition Only)



Notes:

Figure B.2: Heterogeneity: Ratings by Employers' Diversity Preference - Additional Outcomes (Non-blind Condition Only)

B.I Does treatment effect vary by resume order?

To assess whether evaluators lose focus as the survey progresses (and to verify that our main effects are not driven by differential attention at later profile positions), we extend equation (2) by allowing the group-specific blinding effect to vary flexibly with the resume's position in respondent s's evaluation sequence, $p_{i,s} \in \{1, \ldots, 18\}$. In particular, we estimate:

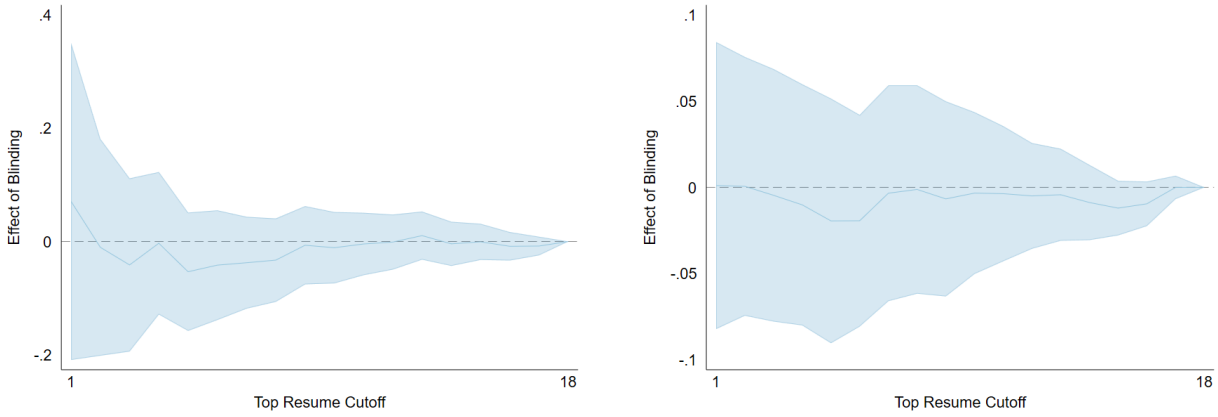
$$y_{i,s} \;=\; \gamma_0 - \sum_{d \in G} \sum_{p=1}^{18} \gamma_{d,p}\, \mathbb{1}\{\texttt{Race_Gender}_{i,s} \;=\; d\}\, \mathbb{1}\{p_{i,s} \;=\; p\} + \mu_i + \omega_{p_{i,s}} + X_s'\beta + \varepsilon_{i,s}, \quad (5)$$

where $\texttt{Race_Gender}_{i,s} \in \{\texttt{Blind}\} \cup G$ and $\texttt{Blind}$ is the omitted base category as in equation (2). The coefficient $\gamma_{d,p}$ can be interpreted as the treatment effect of blinding for group d at sequence position p: $\gamma_{d,p} = \mathbb{E}[y_{i,s} \mid \texttt{Race_Gender}_{i,s} = \texttt{Blind},\, p_{i,s} = p] - \mathbb{E}[y_{i,s} \mid \texttt{Race_Gender}_{i,s} = d,\, p_{i,s} = p]$. If respondents become less attentive over the course of the survey, we would expect the magnitude of $\gamma_{d,p}$ to attenuate toward zero at larger p. Instead, the estimated $\{\gamma_{d,p}\}_{p=1}^{18}$ are stable across p, providing evidence against systematic attention decay and suggesting that the effect of blinding persists throughout the evaluation sequence.

We conduct an analogous check for the role of productivity signals Z_i that are constant within resume profiles like GPA or test cases passed. Because these characteristics are fixed at the resume level, we do not include resume fixed effects in this exercise. We run the following interaction regression:

$$\begin{aligned} y_{i,s} = \alpha_s + \delta_g + \omega_{p_{i,s}} + \beta\, \texttt{Blind}_{i,s} + \sum_{p=1}^{18} \gamma_p(\omega_{p_{i,s}} \times \texttt{Blind}_{i,s}) + \theta Z_i + \phi\,(Z_i \times \texttt{Blind}_{i,s}) \\ + \sum_{p=1}^{18} \psi_p(\omega_{p_{i,s}} \times X_i) + \sum_{p=1}^{18} \kappa_p(\omega_{p_{i,s}} \times X_i \times \texttt{Blind}_{i,s}) + \varepsilon_{i,s}, \quad (6) \end{aligned}$$

The coefficient of interest that we plot is $\Delta_p = \phi + \kappa_p$. The resulting series $\{\Delta_p\}_{p=1}^{18}$ is close to constant across p for both GPA (HR and engineers) and test cases passed (engineers), indicating that evaluators' reliance on these resume-level signals does not systematically change over the survey sequence. Together, these order-heterogeneity checks support an interpretation of our main estimates as reflecting stable decision rules rather than shifting attention or fatigue over the course of the task.



(a) HR

(b) Engineer

Figure B.3: Treatment Effect on Actual Coding Score

Figure B.4: Treatment Effect of Blinding by Resume Order - HR

Figure B.5: Treatment Effect of Blinding by Resume Order - Engineer

Figure B.6: Productivity Signals by Resume Order

B.II Does previous resume characteristics affect future evaluation?

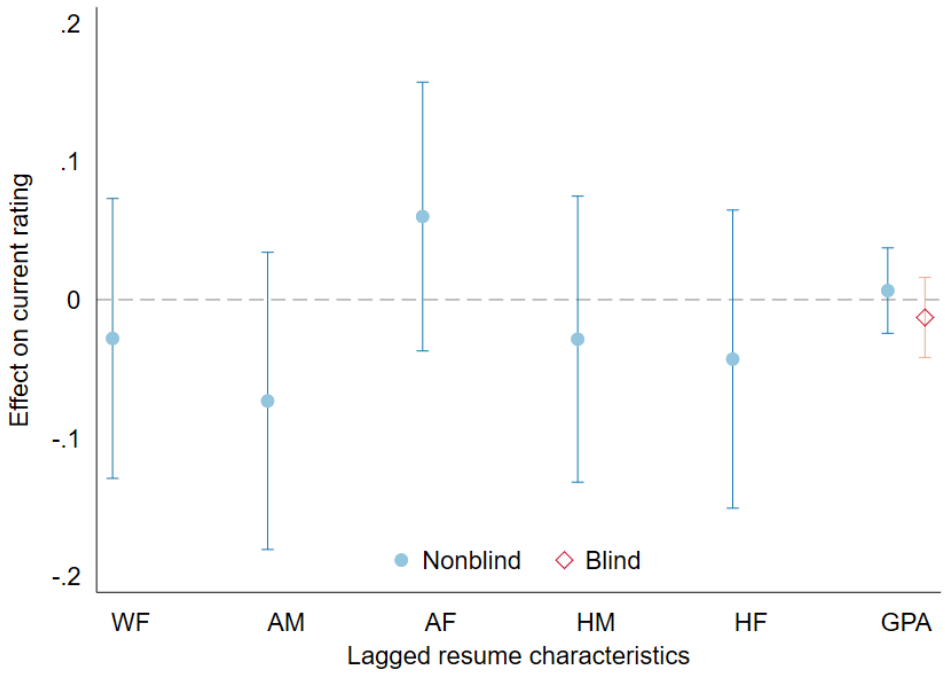


Figure B.7: Effects of Previous Resume on Current Evaluation

C Experiment Details

C.I Experiment: Applicant Phase

C.I.1 Recruitment - Student Participants

Sample Recruitment Email (UC Berkeley CS / EECS Students)

We, a group of PhD students, are looking for student participants for our research project (IRB approved, HUM00242325) that studies the hiring process for early-career engineering roles in the U.S. tech industry. Please review the information below. We hope many of you will consider participating.

Why should you participate in our study?

1. **Monetary compensation**
   - $20 gift card upon completing the coding task, providing your resume, and filling out the survey.
   - Up to an additional $20 based on coding task performance.
2. **A low-stakes chance to practice coding interview tasks**
3. **Visibility to potential employers**
   - With your permission, your resume will be shared (as part of our experiment) with recruiters, HR professionals, and engineers at a diverse set of companies.
   - While this is not a formal job application channel, it may lead to referrals or invitations to apply. A prerequisite for firm participants is that they are part of HR or engineering teams currently recruiting CS graduates.
   - Our goal is to make the study valuable for students—as practice, feedback, and/or exposure to potential employers.

Who is eligible?

- 3rd- and 4th-year students in CS / EECS.

What would you have to do?

- **Provide your most recent resume (electronic version).**
  - If you agree, we will share your resume with participating tech employers who are actively seeking CS graduates. Employers may contact you to refer you or invite you to apply and/or interview. It is entirely up to you whether to engage.
- **Complete coding tasks (total: 1 hour).**
  - Tasks are designed by experienced industry professionals to resemble coding interview questions.
  - You will have one hour to complete the tasks at XLab.
- **Complete a brief 10-minute survey.**

How can I participate?
If you are interested, please fill out this RSVP survey. If you meet eligibility criteria, we will provide detailed information about the study: RSVP Survey Link.

Contact information:
For any inquiries, feel free to contact (PI).

Thank you for considering participation in our study!

C.I.2 Coding Questions

Figures C.1 and Figure C.2 show the two coding questions that were administered to the students. These questions were taken from LeetCode under the guidance of two professional tech industry software engineers. To mitigate the concerns of leakage (students sharing details of our lab with those taking the lab at a later date), we manually changed some of the text to make the context different, while keeping the same core question. Engineers in the employer phase of the experiments were told that there may be some variation in the coding questions.

Assume you are an awesome parent and want to give your children some cookies. But, you should give each child at most one cookie.

Each child `i` has a greed factor `g[i]`, which is the minimum size of a cookie that the child will be content with; and each cookie `j` has a size `s[j]`. If `s[j] >= g[i]`, we can assign the cookie `j` to the child `i`, and the child `i` will be content. Your goal is to maximize the number of your content children and output the maximum number.

Example 1:

```
Input: g = [1,2,3], s = [1,1]
Output: 1
Explanation: You have 3 children and 2 cookies. The greed factors of 3 children are 1, 2, 3.
And even though you have 2 cookies, since their size is both 1, you could only make the child whose
greed factor is 1 content.
You need to output 1.
```

Example 2:

```
Input: g = [1,2], s = [1,2,3]
Output: 2
Explanation: You have 2 children and 3 cookies. The greed factors of 2 children are 1, 2.
You have 3 cookies and their sizes are big enough to gratify all of the children,
You need to output 2.
```

Constraints:

- $1 <= g.length <= 3 * 10^4$
- $0 <= s.length <= 3 * 10^4$
- $1 <= g[i], s[j] <= 2^{31} - 1$

Figure C.1: Q1 of the coding task

Given an `m x n` 2D binary grid `grid` which represents a map of `'1'`s (land) and `'0'`s (water), return *the number of islands*.

An **island** is surrounded by water and is formed by connecting adjacent lands horizontally or vertically. You may assume all four edges of the grid are all surrounded by water.

Example 1:

```
Input: grid = [
  ["1","1","1","1","0"],
  ["1","1","0","1","0"],
  ["1","1","0","0","0"],
  ["0","0","0","0","0"]
]
Output: 1
```

Example 2:

```
Input: grid = [
  ["1","1","0","0","0"],
  ["1","1","0","0","0"],
  ["0","0","1","0","0"],
  ["0","0","0","1","1"]
]
Output: 3
```

Constraints:

- `m == grid.length`
- `n == grid[i].length`
- `1 <= m, n <= 300`
- `grid[i][j]` is `'0'` or `'1'`.

Figure C.2: Q2 of the coding task

C.II Experiment: Employer Phase

C.II.1 Recruitment - Firm Participants

Sample Recruitment Email (Software Engineer)

Dear Software Engineer / Name,

We are a research team at UC Berkeley and the University of Michigan studying how companies recruit early-career CS graduates. We'd love your perspective as someone involved in hiring. In a short survey (approx. 20 minutes), we will ask about your insights on how candidates are evaluated and have you review a few student candidate "applications." In return, you'll receive:

- A $40 gift card (Amazon or Target) for completing the survey.
- (Optional) An additional $10 gift card for each colleague you invite from your firm who completes the survey.
- Entries into a lottery for six $50 gift cards.
- (Optional) A personalized list of upcoming UC Berkeley Computer Science graduates vetted via our in-person, unassisted (no AI access) coding tests and matched to your preferences, whom you can consider for hire or referral.

Why this matters
Your input helps us, as academic researchers, understand how hiring decisions are made—and could shape future recruiting practices. You'll also gain early access to qualified CS graduates, complete with résumés, objective code scores (graded by industry professionals), and unassisted coding samples.

This survey is optimized for display on a laptop/desktop; attempts on a phone/tablet will be screened out.

Follow this link to the survey: Take the survey

For questions, reach out anytime at (PI Email and LinkedIn)

Thank you for your time, and we look forward to your insights!

Sincerely,

(PI Signature)

* This program is approved by the University of Michigan and University of California Institutional Review Boards.
** Your email was obtained via an online platform (Apollo.io) that allows users to find targeted industry contacts.
Opt out of future emails: Click here to unsubscribe.

C.II.2 Resume Creation and Name Selection

We will reformat the student resumes collected from the UCB CS students who participated in the coding task in order to control for non-quantifiable variations in resume designs and to focus on variation in quantity and quality of students' academic, leadership, and career-oriented (work and project) experiences. The reformatted resume closely resembles the actual resumes provided by the student participants - only occasional and minor changes are made in order to remove obvious gender and race/ethnicity signals from students' descriptions of experiences. For example, a description of extra-curricular activity organizing a club trip to Chicago for Chinese students would

have the word "Chinese" removed, such that the gist of responsibility and accomplishment would not change, while allowing for the blinding to remain intact when this resume is evaluated by the firms assigned to the full and partial blinding treatment arms.

To create the names for the resumes, we followed the approach by **?**. For first names, we select the most common names in California between 2000-2007 using data from the Social Security Administration. We only keep the first names that strongly signal gender, by keeping the first names that have more than 99% occurrence in one gender. We then used (Tzioumis, 2018) that reports the probability of race for each first names, and kept the names that had probability greater than 70% for each race.[17] We also removed any names that strongly signals religion, or had names that could be ambiguous in race/gender from outside-US context.[18] For the last names, we used the data from Census 2010 and kept all last names that has probability greater than 70% to belong to a specific race. We then took the most common last names within each race, and randomly matched the last names with the first names within the same race to create the full names.[19]

C.II.3 Embedded On-Task Tracker

Both stages of the employer-side survey embed "TaskMaster" (Permut, 2019) in Qualtrics to record, for each page, *active time* and total elapsed time. This distinguishes time actually spent on the page from time when the survey tab is background.[20] Therefore, TaskMaster may provide better measure of time spent on evaluating each candidate and also provides indirect evidence of participants *potentially* consulting outside sources (i.e. Google / LinkedIn searches) that could undo our randomized and assigned demographic signal by revealing true underlying demographics. Note, in the survey, at the end of Module, we do ask firm participants directly whether they used outside sources and searched for the resume owners, while evaluating the candidates.

C.II.4 Stage 1: Baseline Employer Survey for HR professionals

The first stage of the employer-side experiment resembles the *resume-screening* stage of a hiring process, where employers conduct an initial screening based on information revealed in the resumes and decide whether to pass the resume to a subsequent technical interview stage. The survey consists of several modules, including the randomization stage.

Module 1. Consent and Screening: Baseline employer (HR) survey begins with consent and screening questions. The screening criteria are built as follows:

1. Employers who consent to participate in the study

[17] For Asian first name, we used the same names as the White names, because there is no popular Asian first names. Race is signaled by last name for Asian.

[18] Christian and Jesus is removed for religious names, and names like Andrea which is feminine in US but masculine in other contexts are removed.

[19] For Asian last names, we chose Chinese and Indian last names, because we are also interested in the differences in employers' evaluation of Chinese and Indian applications, two of the most common races in the tech industry.

[20] Qualtrics reports total elapsed time for the survey page; TaskMaster can exclude off-tab intervals.

2. Employers who have not participated in the study before.
3. Employers whose employment status is full-time or part-time employees.
4. Employers who reside and work in the U.S.
5. Employers who are aged 18 or over.
6. Employers who consider themselves tech industry workers.
7. Employers who consider themselves HR professionals.
8. Employers who have experience hiring for technical positions.

Employers who do not satisfy the above conditions are screened out.

Module 2. Demographic Questions: We collect information on their demographic characteristics, firm characteristics, and information on the position the company is hiring for.

Module 3. Randomization and Candidate Evaluation : Firm participants are randomized into control and treatment groups. Participants assigned to the treatment group receive resumes with applicants' names blinded (see Figure C.4), while participants in the control group receive the same set of resumes, but the names of the applicants will be revealed (see Figure C.3). Note that the names on the resumes are not real student names, but hypothetical names, which are randomly assigned to each resume (see more details about resume creation in Section C.II.2).

After being randomized, firm participants are asked to evaluate 18 student resumes across various dimensions:

1. How good do you think this person will be in *coding*? [scale from 1-10][21]
2. How good of a *fit* do you think this person will be in your workplace? [scale from 1-10]
3. How likely would this person *stay* in your company for the next 5 years? [scale from 1-10]
4. How would you rate this person *overall*? [score between 0 and 100]
5. Would you *select* this person *to advance* to the next hiring stage for more in-depth technical evaluation? (yes / no)

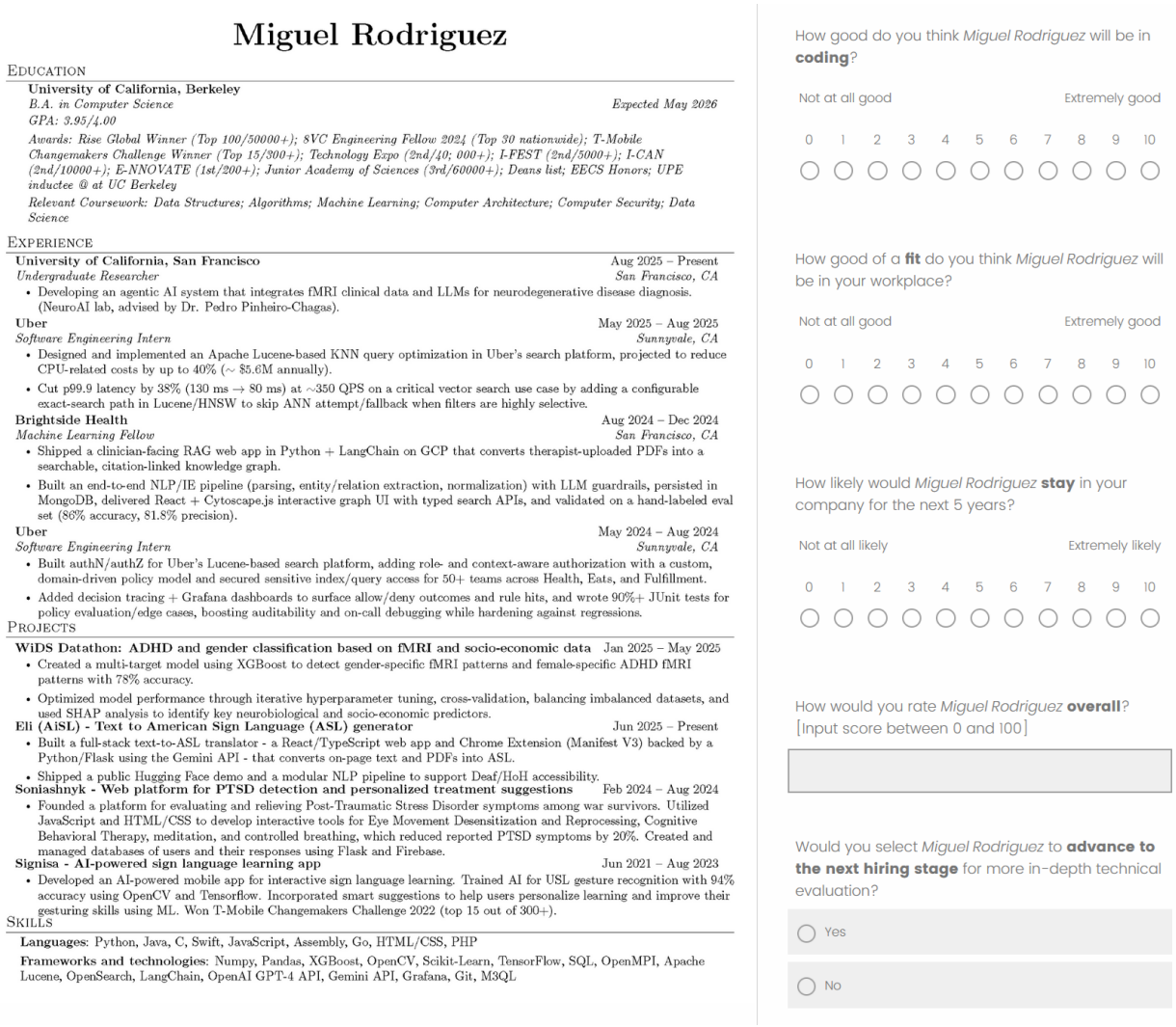
Module 4. Questions about Hiring Experience: We ask about participants' past hiring experiences, such as the type of hiring stages that they have been involved in or how long they have been involved in recruiting for technical positions. We ask questions about how their firms conduct the hiring process and what factors are important in assessing a job applicant's productivity.

Module 5. Questions about Current Hiring Policies: We collect information on the hiring policies of the firm where participants work and ask whether the firms implement any policies related to diversity, equity, and inclusion (DEI) during hiring process and their views on DEI policies.

[21] Participants are incentivized to correctly guess the student's actual coding ability via a lottery.

Module 6. Invitation of co-workers and IRR: Finally, participants are asked to invite HR or engineer from the same firm.

Before exiting the survey, participants are asked whether they would like to be connected with UC Berkeley's CS students. If they say "yes", we email them real resumes and codes of students that correspond to their own preferences revealed in the survey, as in the IRR method.



Miguel Rodriguez

EDUCATION

University of California, Berkeley
B.A. in Computer Science — *Expected May 2026*
GPA: 3.95/4.00

Awards: Rise Global Winner (Top 100/50000+); 8VC Engineering Fellow 2024 (Top 30 nationwide); T-Mobile Changemakers Challenge Winner (Top 15/300+); Technology Expo (2nd/40; 000+); I-FEST (2nd/5000+); I-CAN (2nd/10000+); E-NNOVATE (1st/200+); Junior Academy of Sciences (3rd/60000+); Deans list; EECS Honors; UPE inductee @ at UC Berkeley

Relevant Coursework: Data Structures; Algorithms; Machine Learning; Computer Architecture; Computer Security; Data Science

EXPERIENCE

University of California, San Francisco — Aug 2025 – Present
Undergraduate Researcher — *San Francisco, CA*
- Developing an agentic AI system that integrates fMRI clinical data and LLMs for neurodegenerative disease diagnosis. (NeuroAI lab, advised by Dr. Pedro Pinheiro-Chagas).

Uber — May 2025 – Aug 2025
Software Engineering Intern — *Sunnyvale, CA*
- Designed and implemented an Apache Lucene-based KNN query optimization in Uber's search platform, projected to reduce CPU-related costs by up to 40% (∼ $5.6M annually).
- Cut p99.9 latency by 38% (130 ms → 80 ms) at ∼350 QPS on a critical vector search use case by adding a configurable exact-search path in Lucene/HNSW to skip ANN attempt/fallback when filters are highly selective.

Brightside Health — Aug 2024 – Dec 2024
Machine Learning Fellow — *San Francisco, CA*
- Shipped a clinician-facing RAG web app in Python + LangChain on GCP that converts therapist-uploaded PDFs into a searchable, citation-linked knowledge graph.
- Built an end-to-end NLP/IE pipeline (parsing, entity/relation extraction, normalization) with LLM guardrails, persisted in MongoDB, delivered React + Cytoscape.js interactive graph UI with typed search APIs, and validated on a hand-labeled eval set (86% accuracy, 81.8% precision).

Uber — May 2024 – Aug 2024
Software Engineering Intern — *Sunnyvale, CA*
- Built authN/authZ for Uber's Lucene-based search platform, adding role- and context-aware authorization with a custom, domain-driven policy model and secured sensitive index/query access for 50+ teams across Health, Eats, and Fulfillment.
- Added decision tracing + Grafana dashboards to surface allow/deny outcomes and rule hits, and wrote 90%+ JUnit tests for policy evaluation/edge cases, boosting auditability and on-call debugging while hardening against regressions.

PROJECTS

WiDS Datathon: ADHD and gender classification based on fMRI and socio-economic data — Jan 2025 – May 2025
- Created a multi-target model using XGBoost to detect gender-specific fMRI patterns and female-specific ADHD fMRI patterns with 78% accuracy.
- Optimized model performance through iterative hyperparameter tuning, cross-validation, balancing imbalanced datasets, and used SHAP analysis to identify key neurobiological and socio-economic predictors.

Eli (AiSL) - Text to American Sign Language (ASL) generator — Jun 2025 – Present
- Built a full-stack text-to-ASL translator - a React/TypeScript web app and Chrome Extension (Manifest V3) backed by a Python/Flask using the Gemini API - that converts on-page text and PDFs into ASL.
- Shipped a public Hugging Face demo and a modular NLP pipeline to support Deaf/HoH accessibility.

Soniashnyk - Web platform for PTSD detection and personalized treatment suggestions — Feb 2024 – Aug 2024
- Founded a platform for evaluating and relieving Post-Traumatic Stress Disorder symptoms among war survivors. Utilized JavaScript and HTML/CSS to develop interactive tools for Eye Movement Desensitization and Reprocessing, Cognitive Behavioral Therapy, meditation, and controlled breathing, which reduced reported PTSD symptoms by 20%. Created and managed databases of users and their responses using Flask and Firebase.

Signisa - AI-powered sign language learning app — Jun 2021 – Aug 2023
- Developed an AI-powered mobile app for interactive sign language learning. Trained AI for USL gesture recognition with 94% accuracy using OpenCV and Tensorflow. Incorporated smart suggestions to help users personalize learning and improve their gesturing skills using ML. Won T-Mobile Changemakers Challenge 2022 (top 15 out of 300+).

SKILLS

Languages: Python, Java, C, Swift, JavaScript, Assembly, Go, HTML/CSS, PHP

Frameworks and technologies: Numpy, Pandas, XGBoost, OpenCV, Scikit-Learn, TensorFlow, SQL, OpenMPI, Apache Lucene, OpenSearch, LangChain, OpenAI GPT-4 API, Gemini API, Grafana, Git, M3QL

How good do you think *Miguel Rodriguez* will be in **coding**?

Not at all good — Extremely good

0 1 2 3 4 5 6 7 8 9 10

How good of a **fit** do you think *Miguel Rodriguez* will be in your workplace?

Not at all good — Extremely good

0 1 2 3 4 5 6 7 8 9 10

How likely would *Miguel Rodriguez* **stay** in your company for the next 5 years?

Not at all likely — Extremely likely

0 1 2 3 4 5 6 7 8 9 10

How would you rate *Miguel Rodriguez* **overall**? [Input score between 0 and 100]

Would you select *Miguel Rodriguez* to **advance to the next hiring stage** for more in-depth technical evaluation?

Yes

No

Figure C.3: Screenshot of Employer Survey for HR: Resume Screening in Non-Blind Condition

[redacted]

EDUCATION

University of California, Berkeley
B.A. in Computer Science — *Expected May 2026*
GPA: 3.95/4.00
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Uber — May 2024 – Aug 2024
Software Engineering Intern — *Sunnyvale, CA*
- Built authN/authZ for Uber's Lucene-based search platform, adding role- and context-aware authorization with a custom, domain-driven policy model and secured sensitive index/query access for 50+ teams across Health, Eats, and Fulfillment.
- Added decision tracing + Grafana dashboards to surface allow/deny outcomes and rule hits, and wrote 90%+ JUnit tests for policy evaluation/edge cases, boosting auditability and on-call debugging while hardening against regressions.

PROJECTS

WiDS Datathon: ADHD and gender classification based on fMRI and socio-economic data — Jan 2025 – May 2025
- Created a multi-target model using XGBoost to detect gender-specific fMRI patterns and female-specific ADHD fMRI patterns with 78% accuracy.
- Optimized model performance through iterative hyperparameter tuning, cross-validation, balancing imbalanced datasets, and used SHAP analysis to identify key neurobiological and socio-economic predictors.

Eli (AiSL) - Text to American Sign Language (ASL) generator — Jun 2025 – Present
- Built a full-stack text-to-ASL translator - a React/TypeScript web app and Chrome Extension (Manifest V3) backed by a Python/Flask using the Gemini API - that converts on-page text and PDFs into ASL.
- Shipped a public Hugging Face demo and a modular NLP pipeline to support Deaf/HoH accessibility.

Soniashnyk - Web platform for PTSD detection and personalized treatment suggestions — Feb 2024 – Aug 2024
- Founded a platform for evaluating and relieving Post-Traumatic Stress Disorder symptoms among war survivors. Utilized JavaScript and HTML/CSS to develop interactive tools for Eye Movement Desensitization and Reprocessing, Cognitive Behavioral Therapy, meditation, and controlled breathing, which reduced reported PTSD symptoms by 20%. Created and managed databases of users and their responses using Flask and Firebase.

Signisa - AI-powered sign language learning app — Jun 2021 – Aug 2023
- Developed an AI-powered mobile app for interactive sign language learning. Trained AI for USL gesture recognition with 94% accuracy using OpenCV and Tensorflow. Incorporated smart suggestions to help users personalize learning and improve their gesturing skills using ML. Won T-Mobile Changemakers Challenge 2022 (top 15 out of 300+).

SKILLS

Languages: Python, Java, C, Swift, JavaScript, Assembly, Go, HTML/CSS, PHP

Frameworks and technologies: Numpy, Pandas, XGBoost, OpenCV, Scikit-Learn, TensorFlow, SQL, OpenMPI, Apache Lucene, OpenSearch, LangChain, OpenAI GPT-4 API, Gemini API, Grafana, Git, M3QL

How good do you think this person will be in **coding**?

Not at all good — Extremely good

0 1 2 3 4 5 6 7 8 9 10

How good of a **fit** do you think this person will be in your workplace?

Not at all good — Extremely good

0 1 2 3 4 5 6 7 8 9 10

How likely would this person **stay** in your company for the next 5 years?

Not at all likely — Extremely likely

0 1 2 3 4 5 6 7 8 9 10

How would you rate this person **overall**? [Input score between 0 and 100]

Would you select this person to **advance to the next hiring stage** for more in-depth technical evaluation?

Yes

No

Figure C.4: Screenshot of Employer Survey for HR: Resume Screening in Blind Condition

Note: In the Blind condition, engineers evaluate the candidates with names (demographic signal) redacted on the resume and in the questions.

C.II.5 Stage 2: Baseline Employer Survey for Software Engineers

The second stage with technical professionals resembles the *technical interview* stage of a job application, where employers gain more information about job applicants by having them complete specific coding tasks. All modules of the survey are identical to the HR professionals employer survey described in section C.II.4, except module 3.

In module 3, technical professionals will see the same set of resumes for evaluation and follow the same randomization as HR personnel. In line with the technical evaluation engineering team usually conduct, they will additionally see *the codes of the applicants* collected from our coding task provided with details of the coding task and simple metrics on student performance (whether the code compiles, and *the accuracy score* described in **??**).[22] Technical professionals will evaluate resumes for the same set of questions as the HR, except for the last question that asks the HR if the resumes should be passed to the interview stage.

[22] We do not provide the full coding score graded by the industry professional that additionally comprises readability and design, and efficiency; See **??**.

Show Full Resume

Alejandra Perez

Education

University of California, Berkeley
B.A. in Computer Science — *Expected May 2026*
GPA: 3.67/4.00 | Major GPA: 3.67/4.00
Awards: Dean's List; Spring 2025; Headstarter Fellow; HMC Programs Finalist; Rise Fellow
Relevant Coursework: Machine Learning; Data Science; Computer Security; Data Structures and Algorithms; Computer Architecture; Designing Information Devices and Systems; Embedded Systems; Operating Systems; Software Engineering (CodePath; Headstarter); Linear Algebra; Multivariable Calculus; Discrete Maths and Probability.

Experience

Apple — May 2025 – Aug 2025
Software Engineering Intern — *Cupertino, CA*
- Designed and implemented a runtime feature-flag framework for AirPods that lets teams toggle and tune features without rebuilds, with ~10% faster iteration in testing, pending rollout.
- Engineered the cross-repo groundwork (5 repos across iOS UI, Bluetooth stack, accessory runtime, and tooling) that enabled runtime flags, introducing a typed config schema, stable APIs, and validation rules to prevent invalid states.
- Coordinated with 5+ teams (firmware, Bluetooth, audio, and QA) to plan the UX and interfaces. Wrote the design doc specifying flag taxonomy, scopes/defaults, API contracts, and failure behavior.

Uber — May 2024 – Aug 2024
Software Engineering Intern — *Sunnyvale, CA*

In our coding test with two questions below:
Time used: 53 / 60 minutes

Question 1 "Assigning Cookies"

Click to View Question

Show this student's full code

- Number of Test Cases Solved: 9/9
- Compile? Yes

```
import sys
import ast
from typing import List


def getContentChildren(g: List[int], s: List[int]) -> int:
    g = sorted(g, reverse=True)
    s = sorted(s)

    content = 0
    for greed in g:
      if not s:
        break
      if greed <= s[-1]:
```

Question 2 "Number of Islands"

Click to View Question

Show this student's full code

- Number of Test Cases Solved:9/10
- Compile? Yes

```
import sys
import ast

# can be optimized both in terms of time and space complexity
# by using set (only 1 set would suffice)

# this interpreter says list is unhashable when trying to add it to a set

coords = [] # all coordinates of 1's
visited = []

def explore(i, j, m, n):
  visited.append([i, j])
  # the following can be made more efficient but is easy to read and debug
```

How good do you think *Alejandra Perez* will be in **coding**?

Not at all good — Extremely good

0 1 2 3 4 5 6 7 8 9 10

How good of a **fit** do you think *Alejandra Perez* will be in your workplace?

Not at all good — Extremely good

0 1 2 3 4 5 6 7 8 9 10

How likely would *Alejandra Perez* **stay** in your company for the next 5 years?

Not at all likely — Extremely likely

0 1 2 3 4 5 6 7 8 9 10

How would you rate *Alejandra Perez* **overall**? [Input score between 0 and 100]

Figure C.5: Screenshot of Employer Survey for Engineer: (Technical) Interview in Non-Blind Condition

Show Full Resume

[redacted]

EDUCATION

University of California, Berkeley
B.A. in Computer Science — *Expected May 2026*
GPA: 3.84/4.00
Relevant Coursework: Data Structures; Algorithms; Database Systems (SQL); Computer Security; Machine Learning; Introduction to the Internet; Robotics (ROS); Computer Vision; Designing Information Devices and Systems; Operating Systems and System Programming

EXPERIENCE

Oscr AI — Jun 2025 – Aug 2025
Software Engineer Intern — *Remote*
- Architected an end-to-end SEO content generation platform using FastAPI and React (TypeScript), cutting draft-to-publish time by ~70% for pilot users.
- Built nSn GEO (Generative Engine Optimization) pipelines with AI agents using prompting and RAG, automating article drafting, enrichment, and streamline publishing.
- Implemented authentication with Supabase for authentication testing, validating login-gated flows while avoiding exposure of real user data.

Ochy — Jan 2024 – May 2024
Software Engineer Intern — *Remote*

In our coding test with two questions below:
Time used: 39 / 60 minutes

Question 1 "Assigning Cookies"
Click to View Question
Show this student's full code

- Number of Test Cases Solved: 9/9
- Compile? Yes

```
import sys
import ast
from typing import List


def getContentChildren(g: List[int], s: List[int]) -> int:
    # your solution
    # return the max number of the content children
    #sorted each arr, so we move the ptr in g, if the item is smaller
    #than s[ptr], add 1 to return val, else, we move ptrs
    if not g or not s:
      return 0
    res = 0
    g = sorted(g)
```

Question 2 "Number of Islands"
Click to View Question
Show this student's full code

- Number of Test Cases Solved:9/10
- Compile? Yes

```
import sys
import ast

def numIslands(grid):
    # your solution here
    # get a grid, output the num of island
    # start from (0, 0), then using dfs to visit all item ==1, mark as visited by using isl
    # then go to (0, 1) if == 1, check if visited, if not run dfs again.

    res = 0
    temp = []
    visited = []
    row, col = len(grid), len(grid[0])
```

How good do you think this person will be in **coding**?

Not at all good — Extremely good

0 1 2 3 4 5 6 7 8 9 10

How good of a **fit** do you think this person will be in your workplace?

Not at all good — Extremely good

0 1 2 3 4 5 6 7 8 9 10

How likely would this person **stay** in your company for the next 5 years?

Not at all likely — Extremely likely

0 1 2 3 4 5 6 7 8 9 10

How would you rate this person **overall**? [Input score between 0 and 100]

Figure C.6: Screenshot of Employer Survey for Engineer: (Technical) Interview in Blind Condition

Note: In the Blind condition, engineers evaluate the candidates with names (demographic signal) redacted on the resume and in the questions.

C.II.6 Follow-Up Survey

We will conduct a short follow-up employer survey within 6 months after our baseline survey for those who participated in our study and received students' resumes. The aim is to obtain the following information:

- Whether they reached out to any of the candidates we matched them with.
- Whether any candidate(s) proceeded to further stages (e.g., interview) or were eventually hired.
- The reasons behind not reaching out to candidates at all (if applicable).

This information will be utilized to assess the effectiveness of the experiment's incentives related to the IRR method and to determine the satisfaction of the firms with the recommendations provided.

C.III Data Quality Checks

For the employer survey, although we believe that the incentives designed via the employer-employee matching will be sufficient for employers to report truthfully in the survey, we will include a question aimed at capturing respondents' attention within the survey. We will indicate:

1. Whether employers answered the attention question correctly.
2. Whether employers rank among the top 5% in terms of the fastest total time spent on the survey.
3. Whether employers paid attention to resume evaluation.[23]

We will define different samples based on these quality cutoffs and assess whether the quality of submissions is affected by our treatments. We will also collect employers' work email addresses to identify and eliminate duplicate survey responses.

[23] Examples of employers who did not pay attention include those who provided resume evaluations where there is no variation in ratings across all 24 resumes.